

Damping versus oscillations in the setting of Hadžić, Rein, Schrecker and Straub

Loop gain, linear theory and particle-in-cell runs

Working notes

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Abstract

Hadžić, Rein, Schrecker and Straub proved that, in the radial Vlasov–Poisson system with a central point mass and all particles at the same angular momentum, small steady states $\varepsilon(E_t - E)^k$ damp their linear perturbations if $k > 1$ and do not if $1/2 < k \leq 1$, because a discrete eigenvalue appears below the band of orbital frequencies. The theorem holds below a mass $\varepsilon_0(k)$ that it does not estimate. We reproduce their setting with our tools and study it at finite mass. The criterion for the discrete mode is that the gain λ_{edge} of the self-consistent loop at the lower edge of the band exceeds one; λ_{edge} is the norm of the Mathur operator of their proof. At small mass ($a_0 = 0.01$) $\lambda_{\text{edge}} \approx 0.02$ for $k > 1$ and diverges for $k \leq 1$, as the theorem requires, but the discrete mode of $k \leq 1$ lies less than 10^{-6} band widths below the edge. The threshold $\lambda_{\text{edge}} = 1$ is reached at $a_0 \approx 0.53$ for $k = 1.25$ and $a_0 \approx 0.93$ for $k = 1.5$; for $k = 2$ and 3 there is no discrete mode even with a shell as massive as the central mass. In these units the coupling parameter η of our earlier work is 1.25 and 2.6 at the two thresholds, and above 3 for $k = 2$: η does not mark the transition, λ_{edge} does. The linearized solver confirms all of this in time. Particle-in-cell runs with 10^4 particles follow linear theory to 1% up to $t \approx 1000$ at small mass; at $a_0 = 1$ they show the undamped mode of $k = 0.75$ and $k = 1$ at the predicted frequency (to about 10^{-3}) and the Landau damping of $k = 2$ at the predicted rate (to 5%). The modes that lie just below the edge ($k = 1.25$ and 1.5) lose amplitude in the runs. A scan in amplitude, grid and number of particles shows that this loss is a finite-amplitude effect: it grows with ε and does not change with Δr or N . With 4×10^4 particles the mode of $k = 1.25$ persists up to $t = 4000$ at the predicted frequency, and with 10^5 particles and $\varepsilon = 0.03$ its amplitude stays within 0.90 – 1.14 of the linear one. The departure of the reference runs from stationarity has two parts: an initial level set by the grid, and a later growth caused by the discreteness of the particles, which four times more particles remove.

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1 Model and theory

1.1 The setting of the theorem

We consider the radial Vlasov–Poisson system in which all particles have the same modulus of angular momentum L_0 , around a point mass $M = 1$ (with $G = 1$),

$$\partial_t F + p_r \partial_r F - \partial_r \Phi_{\text{ef}} \partial_{p_r} F = 0, \quad \Phi_{\text{ef}} = -\frac{1}{r} + \Phi_{\text{self}} + \frac{L_0^2}{2r^2},$$

where Φ_{self} is the potential of the density $4\pi r^2 \rho = 8\pi^2 L_0 \int F dp_r$. This is the system of Hadžić, Rein, Schrecker and Straub [HRSS], with their squared angular momentum $L = L_0^2$. We use $L_0 = 2$, so that the circular orbit of the bare point mass is at $r = 4$ with energy $E_{\text{min}} = -1/8$, and the radial frequency is $\Omega(J) = (J + 2)^{-3}$ without self-gravity.

The steady states are polytropes in the energy,

$$F_{\text{eq}} = A (E_t - E)^k \quad (E < E_t), \quad F_{\text{eq}} = 0 \quad (E \geq E_t),$$

where k is the exponent of the vacuum edge and $E_t = E(J_t)$ is the energy of the orbit of action J_t , the edge of the support. We use $J_t = 0.7$. Without self-gravity this gives $E_t = -0.0686$, within their single-gap condition $-2^{-2/3}/8 = -0.079 < E_t < 0$, which requires $\Omega_{\text{max}} \geq 2\Omega_{\text{min}}$; here $\Omega_{\text{max}}/\Omega_{\text{min}} = 2.46$. The bands of the harmonics $n = 1, 2, \dots$ then overlap, and the only gap in the spectrum is the principal one, $(0, \Omega_{\text{min}})$. The mass a_0 of the shell is measured in units of the central mass; their small parameter ε is proportional to it.

1.2 The theorem

For $k > 1/2$ and ε below a threshold $\varepsilon_0(k)$ [HRSS, Thm. 1.2]:

- if $1/2 < k \leq 1$ the linearized operator has an eigenvalue in the principal gap, so the perturbation does not damp; this includes the King model;
- if $k > 1$ the point spectrum is empty and the field damps in the time-averaged sense $\lim_{T \rightarrow \infty} T^{-1} \int_0^T \|\nabla U\|^2 dt = 0$, without a rate.

The value of $\varepsilon_0(k)$ is not estimated. The proof uses a Birman–Schwinger principle due to Mathur: there is an eigenvalue in the gap if and only if the norm of the Mathur operator M_λ is at least one for some λ in the gap. For $k > 1$ they show $M_\lambda \leq C\varepsilon$, and for $k \leq 1$ that M_λ diverges at the edge of the gap.

1.3 The loop gain

In angle–action variables a mode $\delta F = a(Q, J)e^{-i\omega t}$ obeys, harmonic by harmonic, $(k\Omega - \omega)a_k = kF'_{\text{eq}}(J)\delta\Phi_k(J)$, where $\delta\Phi_k$ is the harmonic of the potential along the orbit J . (Here k is a Fourier index; the edge exponent is also called k , following [HRSS], and the context distinguishes them.) The density sees only the part of δF even in Q , and with $b_k = a_k + a_{-k}$,

$$b_k = R_k \delta\Phi_k, \quad R_k(J; \omega) = \frac{2k^2\Omega F'_{\text{eq}}}{k^2\Omega^2 - \omega^2},$$

while the Poisson equation gives

$$\delta\Phi_k = \sum_{k'} \int M_{kk'}(J, J') b_{k'} dJ',$$

$$M_{kk'} = 4\pi L_0 \iint \cos kQ \cos k'Q' G(r(Q, J), r(Q', J')) dQ dQ',$$

with $G = -1/\max(r, r')$. A mode exists if $MR(\omega)$ has the eigenvalue one. For $F'_{\text{eq}} < 0$ and $\omega < \Omega_{\text{min}}$ this operator is similar to the symmetric positive operator $|R|^{1/2}(-M)|R|^{1/2}$; its largest eigenvalue $\lambda(\omega)$ is the gain of the loop potential $\rightarrow$ density $\rightarrow$ potential, and $\lambda_{\text{edge}} = \lambda(\Omega_{\text{min}})$. Since λ grows with ω and $\lambda(0) < 1$ (Antonov),

$$\text{there is a discrete mode below the band} \iff \lambda_{\text{edge}} > 1,$$

and its frequency ω_d is the root of $\lambda(\omega_d) = 1$. $\lambda(\omega)$ is the norm of the Mathur operator with $\lambda_{\text{Mathur}} = \omega^2$. Near the edge $|R_1| \propto (J_t - J)^{k-1}/(\Omega - \omega)$, so λ_{edge} is finite only for $k > 1$. We write $\delta_d = (\Omega_{\text{min}} - \omega_d)/\Delta\Omega$ for the distance of the mode below the edge in band widths. The derivation is in Section 1.5 of the demo document (`docs/demo_eta/demo_eta.pdf`).

1.4 The coupling parameter η

In earlier work we used $\eta = \delta\Omega/(\Delta\Omega/\bar{\Omega})$, the self-gravitational shift of the mass-weighted mean frequency, $\delta\Omega = (\bar{\Omega} - \bar{\Omega}_0)/\bar{\Omega}_0$, divided by the relative width of the band; $\bar{\Omega}_0$ is the same average for the same distribution without self-gravity. η needs only the equilibrium. Here we compare it with λ_{edge} .

2 Numerical methods

2.1 Equilibria

The equilibria are built by fixed-point iteration (`equilibrio.py` with `--fondo puntual --forma polE`): the angle–action map of the total potential, the table $E(J)$, the polytrope with $E_t = E(J_t)$ recomputed at each iteration, the density and the Poisson equation, until the potential changes by less than 10^{-13} . The iteration converges in 6–42 iterations for $0.01 \leq a_0 \leq 1$. With the bare point mass the map gives $J(E) = 1/\sqrt{-2E} - 2$ to 1.7×10^{-15} and the circular orbit at $r = 4.000015$. The option is neutral for the isochrone used before: the equilibria, initial states and linear solutions of earlier runs are reproduced bit for bit.

2.2 Loop gain and linear solver

$\lambda(\omega)$ is computed by `lambda_borde.py`: $J = J_t(1 - s^2)$ with 200 Gauss–Legendre nodes in s , 128 points in Q , harmonics up to $k = 6$, and the integrals in Q on a radial grid of 2000 nodes, so that $M = 4\pi L_0 PGP^T$. In this setting, doubling the resolution changes λ_{edge} by at most 2×10^{-4} for $k = 2$ and 3×10^{-3} for $k = 1.25$. For $k \leq 1$, λ diverges at the edge and its value on the grid grows with the resolution; the frequency ω_d is converged.

The linearized solver (`lineal.py`) integrates the linear equation in time on 1600×32 nodes in (J, Q) with $\Delta t = 0.5$, from the initial perturbation $\delta F = F_{\text{eq}}(J/J_t)^{3/2} \cos Q$. The observable is

$$h_1(t) = \frac{\sum \delta F B(J) e^{-iQ}}{\sum F_{\text{eq}} B(J)}, \quad B(J) = J^2 e^{-(J-0.35)^2/0.2^2},$$

and $\|\delta\Phi\|$ is the rms value of $\delta\Phi$ over the radii of the shell.

2.3 PIC runs

The PIC code (`VP_PIC`) uses `BGtype = sphere`, a uniform ball of mass 1 and radius 1: for $r > 1$ it is the point mass, and the shells lie at $1.8 < r < 12.2$. Each run has $400 \times 25 = 10^4$ particles on a regular grid in the angle–action variables of the equilibrium, $\Delta r = 0.1$ on $[0, 20]$, the fourth-order Yoshida integrator and $\Delta t = 0.1$ (Courant factor 2). Each perturbed run (D) has a reference run (Z) with the same particles and $\varepsilon = 0$; the perturbation is $(h_{1,D} - h_{1,Z})/\varepsilon$, and the same for $\delta\Phi$. h_1 is computed with the angle–action map of the equilibrium. Table 1 lists the runs.

series	k	a_0	ε	t_{fin} , output	purpose
1	0.75, 1, 1.5, 2	0.01	0.1 (and Z)	8000, every 20	the regime of the theorem
2	0.75, 1, 1.25, 1.5, 2	1	0.1 (and Z)	4000, every 5	bound modes and the threshold
3	0.75, 1, 1.5, 2	0.01	0.5 (Z of series 1)	8000, every 20	a lower floor
4	1.25, 1.5	1	0.03, 0.3 (Z of series 2); 0.1 with $\Delta r/2$ or $4N$ (and Z)	4000, every 5	amplitude, grid and particles (Section 3.6)
5	1.25	1	0.03 (and Z), with 1280×80 particles and $\Delta t = 0.05$	4000, every 5	a pilot with ten times more particles (Section 3.6)

Table 1: The 36 PIC runs. Each takes 220 s to 380 s with four threads, 840 s to 860 s with 4×10^4 particles, and one hour with 1.024×10^5 particles and half the time step (series 5). The energy is conserved to $|\Delta E/E| \leq 1.3 \times 10^{-7}$ at $a_0 = 0.01$ and $\leq 1.8 \times 10^{-5}$ at $a_0 = 1$.

3 Results

3.1 The loop gain and the threshold mass

Table 2 and Figure 1 give λ_{edge} for 36 equilibria.

- **The theorem at small mass.** For $k > 1$, λ_{edge} is proportional to the mass, 0.016–0.025 at $a_0 = 0.01$, as their bound $M_\lambda \leq C\varepsilon$ says. For $k \leq 1$, λ diverges at the edge, as $\ln(1/\delta)$ for $k = 1$ and as a power for $k = 0.75$, so there is always a mode. At small mass it is bound by less than 10^{-6} band widths.
- **The threshold $\varepsilon_0(k)$.** Interpolating $\lambda_{\text{edge}} = 1$, the mode appears at $a_0 \approx 0.53$ for $k = 1.25$ and $a_0 \approx 0.93$ for $k = 1.5$. For $k = 2$ and 3 there is no mode up to $a_0 = 1$, where $\lambda_{\text{edge}} = 0.82$ and 0.74.
- **Their hypotheses.** $\Omega(J)$ is monotonic in all 36 equilibria. The single-gap condition $\Omega_{\text{max}}/\Omega_{\text{min}} \geq 2$ holds up to $a_0 = 0.3$; it is lost at $a_0 = 0.6$ for $k = 0.75$ (1.99) and at $a_0 = 1$ for $k \leq 1.5$ (1.87–1.98).

$k \setminus a_0$	0.01	0.03	0.1	0.3	0.6	1.0
0.75	∞ ($< 10^{-6}$)	∞ (2×10^{-6})	∞ (10^{-4})	∞ (0.004)	∞ (0.027)	∞ (0.075)
1	∞ ($< 10^{-6}$)	∞ ($< 10^{-6}$)	∞ ($< 10^{-6}$)	∞ (0.0002)	∞ (0.0085)	∞ (0.044)
1.25	0.025	0.075	0.240	0.643	1.112 (0.0001)	1.558 (0.019)
1.5	0.018	0.054	0.170	0.448	0.759 (0.0015)	1.050
2	0.016	0.046	0.145	0.371	0.609	0.820
3	0.016	0.047	0.145	0.358	0.567	0.738

Table 2: λ_{edge} and, in parentheses, the distance δ_d of the discrete mode below the edge in band widths, where there is a mode. For $k \leq 1$, $\lambda_{\text{edge}} = \infty$.

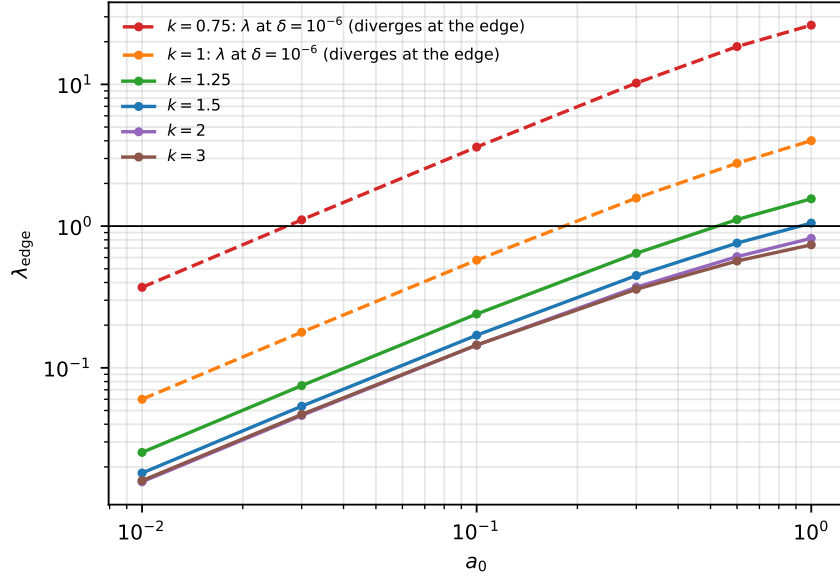


Figure 1: λ_{edge} against the mass for $k > 1$ (solid). For $k \leq 1$, where it diverges, the dashed lines show λ at 10^{-6} band widths below the edge. A discrete mode exists above the horizontal line.

3.2 The coupling parameter η

$k \setminus a_0$	0.01	0.03	0.1	0.3	0.6	1.0
0.75	0.015	0.045	0.160	0.558	1.334	2.668
1	0.016	0.048	0.169	0.586	1.388	2.748
1.25	0.017	0.051	0.178	0.613	1.437	2.818
1.5	0.017	0.053	0.187	0.637	1.481	2.883
2	0.019	0.058	0.202	0.680	1.560	2.997
3	0.022	0.066	0.227	0.751	1.686	3.182

Table 3: η of the same equilibria. The relative width of the band is 0.57–0.80, against 0.16–0.29 in the isochrone of our earlier work.

η is 0.015–0.022 at $a_0 = 0.01$, the regime of the theorem (Table 3). At the threshold it is $\eta \approx 1.25$ for $k = 1.25$ and $\eta \approx 2.6$ for $k = 1.5$, and $\eta = 3.0$ is not enough for $k = 2$. In the isochrone the Wilson model ($k = 2$) had its threshold at $\eta \approx 1.18$. With a band this wide the shape of the edge weighs more in the coupling than the shift of the mean frequency, and η does not mark the transition. The criterion $\lambda_{\text{edge}} = 1$ is the same for every k .

3.3 Linear response in time

At $a_0 = 0.01$ the linear response decays algebraically for every k . The slope of the envelope of $|h_1|$ against $\ln t$ over $1000 \leq t \leq 8000$ is -1.67 , -2.00 , -2.55 and -3.00 for $k = 0.75$, 1 , 1.5 and 2 , close to $t^{-(k+1)}$, the tail of an edge at which the weight vanishes as $(J_t - J)^k$. The sharper the edge, the slower the decay. For $k \leq 1$ the discrete mode is too weakly bound to appear in this time.

At $a_0 = 1$ (Figure 3, black) the solver confirms the loop gain in time. For $k = 0.75$, 1 and 1.25 , $|h_1|$ levels off at 0.075 , 0.071 and 0.059 , and oscillates at 0.14169 , 0.14783 and 0.15330 , against $\omega_d = 0.14169$, 0.14785 and 0.15326 from $\lambda(\omega_d) = 1$. For $k = 1.5$, with the mode 0.0015 band widths below the edge, it decays slowly towards a plateau (0.022 at $t = 4000$) and oscillates at 0.1574 against 0.15751 . For $k = 2$ it is damped: the fitted pole is $\omega = 0.1663$ – 0.1669 , inside the band, with $\gamma = 6.4$ – 6.5×10^{-3} , and $|h_1|$ falls to 10^{-5} by $t = 4000$.

3.4 PIC runs at small mass

The PIC runs follow linear theory to 1 % in $|h_1|$ up to $t \approx 1000$ ($12 \tau_1$, with $\tau_1 = 2\pi/\Delta\Omega = 84$), except $k = 2$ (5 %). After that they reach a floor (Figure 2): the D and Z runs decorrelate, and at $t \approx 8000$ their difference is of the order of the noise of the unperturbed run itself. The floor of $(h_{1,D} - h_{1,Z})/\varepsilon$ at $t \approx 4000$ is $(0.2$ – $1.0) \times 10^{-4}$ with $\varepsilon = 0.1$ and $(1.0$ – $1.3) \times 10^{-4}$ with $\varepsilon = 0.5$. It does not decrease as $1/\varepsilon$, so it is not subtraction noise: the perturbation itself changes how the noise of the particles evolves. To lower it, more particles are needed, not a larger amplitude. At this mass the runs show the damped edge response of every k ; they cannot show the non-damping of $k \leq 1$.

3.5 PIC runs at $a_0 = 1$

Table 4 and Figure 3 give the result.

- **$k = 0.75$ and $k = 1$ do not damp.** The PIC amplitude stays within 10 % of the linear plateau up to $t \approx 1000$, and the frequency agrees with ω_d to 9×10^{-4} and 1.3×10^{-3} (0.14156 against 0.14169 , and 0.14766 against 0.14785).

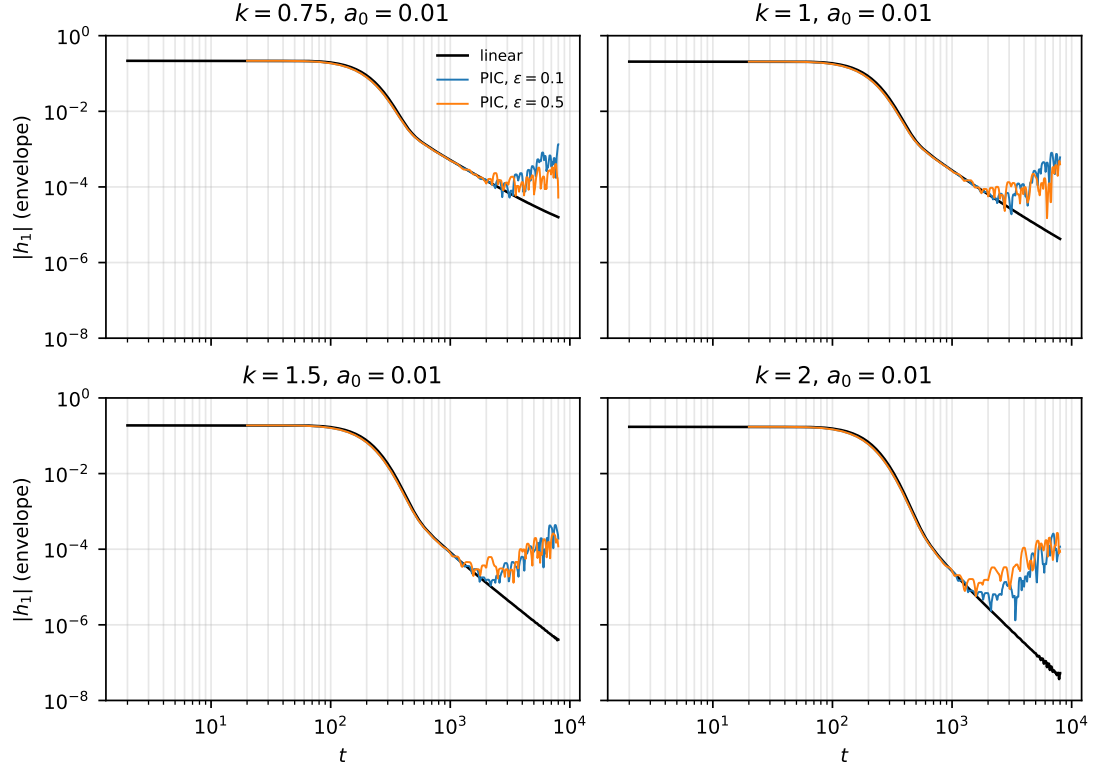


Figure 2: $a_0 = 0.01$: envelope of $|h_1|$ in linear theory (black) and in the PIC runs with $\varepsilon = 0.1$ (blue) and $\varepsilon = 0.5$ (orange).

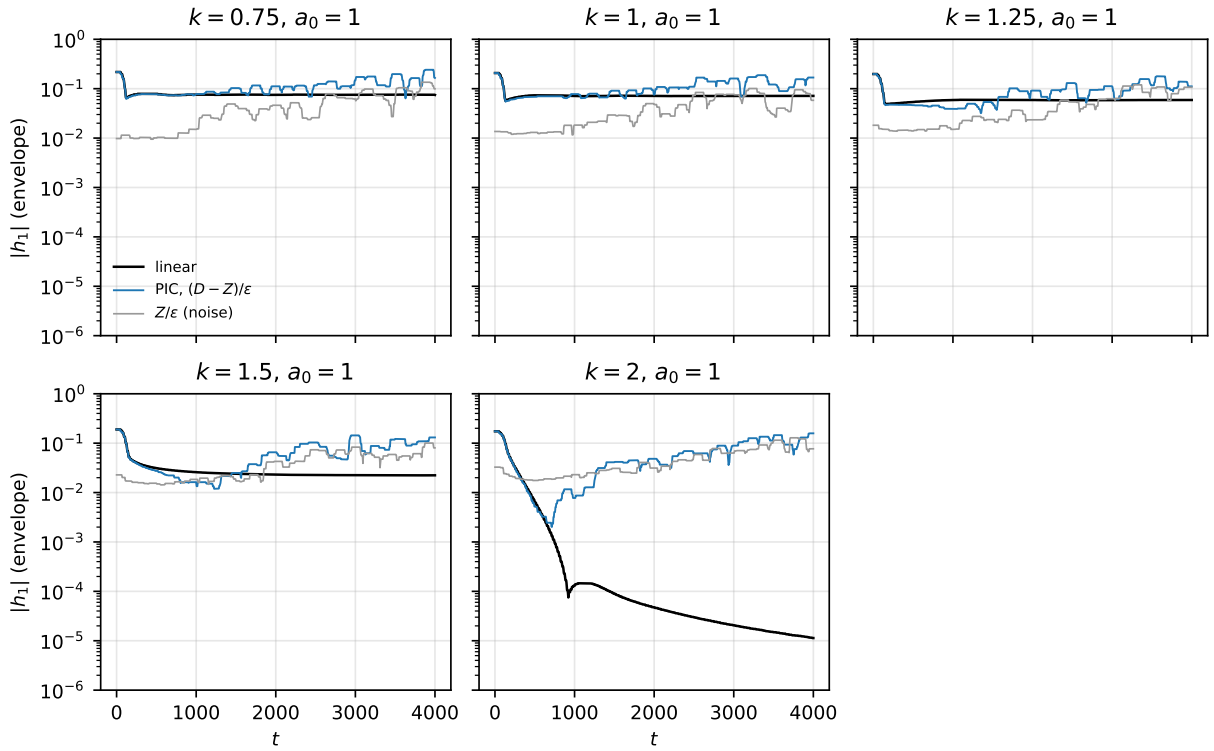


Figure 3: $a_0 = 1$: envelope of $|h_1|$ in linear theory (black), in the PIC runs, $(h_{1,D} - h_{1,Z})/\varepsilon$ (blue), and of the reference run alone, $h_{1,Z}/\varepsilon$ (grey), which measures its departure from stationarity.

k	ω_d ($\lambda = 1$)	window	ω PIC	γ PIC	ω linear	γ linear
0.75	0.14169	300–1500	0.14156	4×10^{-5}	0.14169	4×10^{-6}
1	0.14785	300–1500	0.14766	5×10^{-5}	0.14783	-1×10^{-5}
1.25	0.15326	300–1500	0.15326	4.1×10^{-4}	0.15330	1×10^{-5}
1.5	0.15751	300–1500	0.15805	1.5×10^{-3}	0.15736	1.0×10^{-4}
2	–	100–600	0.16661	6.8×10^{-3}	0.16631	6.5×10^{-3}

Table 4: One-pole fits (matrix pencil, $M = 3$) of h_1 at $a_0 = 1$, before the noise dominates. The fits over 300–1000 (and 100–400 for $k = 2$) give frequencies that differ by at most 5×10^{-4} . For $k = 1.25$ and 1.5 , γ measures a dip of the amplitude, not a decay (Section 3.6).

- **$k = 2$ damps at the linear rate.** The fitted pole agrees with linear theory to 0.2% in ω and 5% in γ , and the PIC amplitude follows the linear one down by a factor of 40, until it reaches the noise at $t \approx 600$.
- **$k = 1.25$ and $k = 1.5$ lose amplitude.** The frequency is that of the discrete mode, but the fitted pole has $\gamma = 4 \times 10^{-4}$ and 1.5×10^{-3} , and the PIC amplitude is 14% to 37% below the linear one at $t = 500$ –1000. These modes lie 0.019 and 0.0015 band widths below the edge. Section 3.6 shows that the loss is a finite-amplitude effect, and that the mode of $k = 1.25$ does not damp.
- **The reference runs are not stationary at this mass.** $|h_{1,Z}|$ grows from 10^{-3} to 10^{-2} between $t = 0$ and 4000, and so does the noise of the subtraction. This limits the comparison to $t \approx 1000$ –1500 (20–40 τ_1 , with $\tau_1 = 37$ –48). Section 3.6 separates the two causes.

3.6 The modes next to the edge

Series 4 tests the loss of the modes of $k = 1.25$ and 1.5 . Each run changes one thing with respect to the run of series 2: the amplitude ($\varepsilon = 0.03$ and 0.3 , with the same reference and the same particles), the grid ($\Delta r = 0.05$ with the same $\Delta t = 0.1$ and the same initial state), or the number of particles ($800 \times 50 = 4 \times 10^4$, which also doubles the resolution of the initial lattice in J and Q). The last two have their own reference. Table 5 gives the PIC amplitude relative to the linear one, and Figure 4 the envelopes.

Series 5 is a pilot for $k = 1.25$ with $1280 \times 80 = 1.024 \times 10^5$ particles, the same ratio of rows in J to points in Q , with $\varepsilon = 0.03$ and $\Delta t = 0.05$, and its own reference. It is the row “10N” of Tables 5 and 6; it is not in Figure 4. It changes the number of particles, the time step and ε at once with respect to series 2, so it does not separate their effects.

- **The loss is a finite-amplitude effect.** At $t = 500$ the PIC amplitude is 0.97, 0.84 and 0.38 of the linear one for $\varepsilon = 0.03$, 0.1 and 0.3 ($k = 1.25$), and 1.02, 0.86 and 0.34 ($k = 1.5$). With $\Delta r/2$ or $4N$ it is 0.84–0.88, the value of $\varepsilon = 0.1$. Over $100 \leq t \leq 500$ the difference $|h_1(\varepsilon) - h_1(0.03)|$ is 3.6–7.1 times larger for $\varepsilon = 0.3$ than for $\varepsilon = 0.1$; a correction to h_1/ε linear in ε gives 3.9 and a quadratic one 9.8. Over the same interval, halving Δr or taking $4N$ changes h_1 by 2.8–13 times less than going from $\varepsilon = 0.03$ to 0.1 . The demo found the same for its mode next to the edge (D3).
- **The mode of $k = 1.25$ does not damp.** With $4N$ the PIC amplitude is 0.6–1.0 of the linear one up to $t = 4000$, and h_1 oscillates at 0.1530 ($\omega_d = 0.15326$). With $\varepsilon = 0.3$ it falls to 0.38 by $t = 500$ and then stays at 0.5–0.9, also at 0.1531. The $\gamma = 4 \times 10^{-4}$ of Table 4 measures a dip of the amplitude, not a decay. With $\varepsilon = 0.1$ the amplitude reaches a minimum of 0.53–0.63 of the linear one at $t \approx 1350$, with 10^4 or 4×10^4 particles and with $\Delta r/2$, and with $4N$ it is back to 1.0 at $t = 2500$. With $\varepsilon = 0.3$ the minimum is deeper

k	run	$t = 500$	1000	2000	3000	frequency, $t \geq 2500$
1.25	$\varepsilon = 0.03$	0.97 (0.91)	1.21 (0.86)	2.79 (1.33)	4.66 (3.63)	0.2210
	$\varepsilon = 0.1$	0.84 (0.27)	0.66 (0.26)	1.55 (0.40)	1.05 (1.09)	0.2179
	$\varepsilon = 0.3$	0.38 (0.09)	0.47 (0.09)	0.69 (0.13)	0.48 (0.36)	0.1531
	$\Delta r/2$	0.88 (0.08)	0.69 (0.23)	1.44 (0.36)	1.88 (1.26)	0.1998
	$4N$	0.84 (0.25)	0.61 (0.24)	0.81 (0.26)	0.99 (0.27)	0.1530
	$10N, \varepsilon = 0.03$	0.98 (0.85)	0.94 (0.79)	0.91 (0.81)	0.91 (0.81)	0.1531
1.5	$\varepsilon = 0.03$	1.02 (1.62)	1.56 (2.56)	6.69 (6.18)	11.2 (12.4)	0.2094
	$\varepsilon = 0.1$	0.86 (0.49)	0.64 (0.77)	2.39 (1.86)	5.98 (3.71)	0.2117
	$\varepsilon = 0.3$	0.34 (0.16)	0.57 (0.26)	1.83 (0.62)	1.34 (1.24)	0.2283
	$\Delta r/2$	0.86 (0.13)	0.80 (0.49)	2.29 (1.18)	3.89 (3.27)	0.2075
	$4N$	0.84 (0.49)	0.41 (0.56)	0.73 (0.63)	1.34 (0.68)	0.1584

Table 5: Envelope of $|h_1|$ (largest value in $[t, t + 100]$) in the PIC runs divided by the linear one, and in parentheses that of the reference run alone, $|h_{1,Z}|/\varepsilon$, divided by the same linear value. The last column is the frequency of h_1 over $2500 \leq t \leq 4000$. In linear theory it is 0.15325 for $k = 1.25$ and 0.15745 for $k = 1.5$; the band starts at $\Omega_{\min} = 0.15597$ and 0.15769.

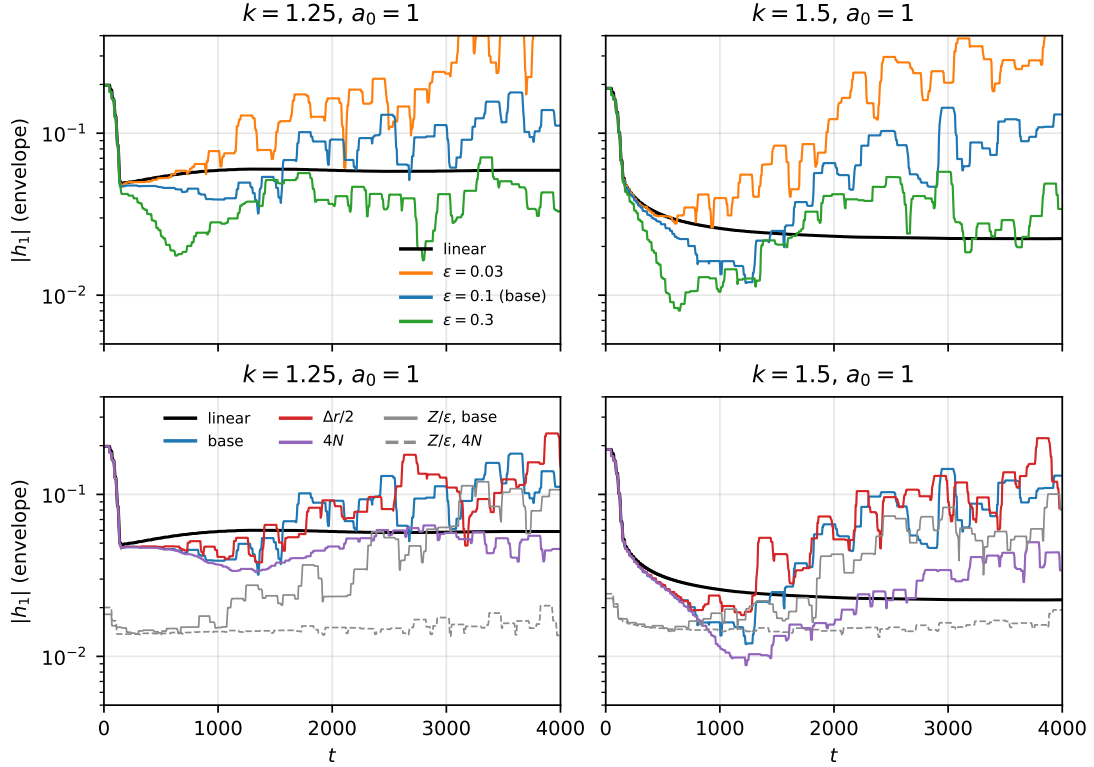


Figure 4: Envelope of $|h_1|$ for $k = 1.25$ (left) and 1.5 (right) at $a_0 = 1$. Top: three amplitudes with the same particles. Bottom: the run of series 2, the same with $\Delta r/2$ and with four times more particles, and the reference runs alone, $|h_{1,Z}|/\varepsilon$ (grey), with 10^4 (solid) and 4×10^4 particles (dashed).

and earlier, 0.32 at $t \approx 630$. In the run of series 2 the noise of the reference reaches the signal at $t \approx 2500$ and hides the recovery.

- **With 10^5 particles the run of $\varepsilon = 0.03$ follows the linear solution up to $t = 4000$.** Its envelope stays between 0.90 and 1.14 of the linear one over $100 \leq t \leq 3900$; with 10^4 particles and the same ε it was 1.2 at $t = 1000$ and 4.7 at $t = 3000$. The minimum, 0.90, is at $t \approx 2250$. With $\varepsilon = 0.1$ it was 0.53–0.63 at $t \approx 1350$ and with $\varepsilon = 0.3$, 0.32 at $t \approx 630$: the dip becomes shallower and later as ε decreases. A pole fitted over $300 \leq t \leq 1000$ and over $300 \leq t \leq 1500$ gives the same frequency, 0.15314, with $|\gamma| \leq 3 \times 10^{-5}$; the linear solution gives 0.15330–0.15332 and the loop gain 0.15326. With 10^4 and 4×10^4 particles the fitted frequency changed with the window (0.15292 and 0.15326; 0.15190 and 0.15325). The difference $\max |h_1 - h_1^{\text{lin}}| / \max |h_1^{\text{lin}}|$ is 0.10 over $500 \leq t \leq 1000$ and 0.23 over $1500 \leq t \leq 2000$, of which 0.07 and 0.12 are in the modulus; a frequency lower by 1.6×10^{-4} shifts the phase by 0.16 and 0.32 at $t = 1000$ and 2000. The reference stays at $|h_{1,Z}| = 1.4 \times 10^{-3}$ from $t = 500$ to $t = 3900$, and the rms displacement of the action of its particles is 0.46, 0.41 and $0.56 \times 10^{-3} J_t$ at $t = 1000$, 2000 and 4000, against 1.4, 6.3 and 37 with 10^4 particles and 0.50, 0.58 and 2.5 with 4×10^4 (`relajacion_N.py`).
- **The mode of $k = 1.5$ is not resolved.** It lies $\Omega_{\min} - \omega_d = 1.8 \times 10^{-4}$ below the edge, so it separates from the response of the edge only after $t \sim 1/(\Omega_{\min} - \omega_d) \approx 5500$, longer than the runs; the linear solution itself still decays at $t = 4000$. With $4N$ the PIC amplitude is 0.4–0.7 of the linear one for $1000 \leq t \leq 2000$ and rises to 1.3 by $t = 3000$, at the frequency 0.1584, while the noise of the reference is 0.6–0.7 of the linear amplitude. That is above $\Omega_{\min}$, but the resolution of a window of 1500 is $2\pi/1500 \approx 4 \times 10^{-3}$. These runs do not decide whether the mode survives.
- **The departure of the references from stationarity has two causes** (Table 6). Up to $t \approx 500$, $|h_{1,Z}|$ is $1.4\text{--}2.3 \times 10^{-3}$; with $\Delta r/2$ it is 2.5–4 times less, and with $4N$ it does not change. This is the error of order Δr^2 of the PIC field, which makes the initial state a slightly wrong equilibrium, as in the demo (its Section 5.2). After $t \approx 1000$, $|h_{1,Z}|$ grows to 10^{-2} by $t = 3900$. With $\Delta r/2$ the growth is the same, and with $4N$ it does not appear up to $t = 4000$. This is the discreteness of the particles, as for M5 in the demo.

k	reference	$t = 0$	500	1000	2000	3000	3900
1.25	$10^4, \Delta r = 0.1$	1.8	1.5	1.5	2.3	6.4	11
	$\Delta r/2$	0.72	0.44	1.4	2.1	7.4	12
	$4N$	2.0	1.4	1.4	1.5	1.6	1.7
	$10N$	2.0	1.4	1.4	1.4	1.4	1.4
1.5	$10^4, \Delta r = 0.1$	2.3	1.5	2.0	4.3	8.3	10
	$\Delta r/2$	0.76	0.40	1.2	2.7	7.3	9.4
	$4N$	2.4	1.5	1.4	1.4	1.5	1.9

Table 6: $|h_{1,Z}|$ of the reference runs at $a_0 = 1$, in units of 10^{-3} (largest value in $[t, t + 100]$).

4 Discussion

Established.

- The criterion $\lambda_{\text{edge}} > 1$, the norm of the Mathur operator of [HRSS] at the edge of the gap, reproduces their dichotomy at small mass: $\lambda_{\text{edge}} \propto a_0$ for $k > 1$ and $\lambda_{\text{edge}} = \infty$ for $k \leq 1$.

- At finite mass it gives the threshold that the theorem does not estimate: $a_0 \approx 0.53$ for $k = 1.25$, $a_0 \approx 0.93$ for $k = 1.5$, above 1 for $k \geq 2$.
- The time-domain linear solver agrees with the loop gain: plateaus at the frequency ω_d , to 4×10^{-5} for $k \leq 1.25$, where $\lambda_{\text{edge}} > 1$, and damping where it is below one.
- The PIC runs reproduce the undamped modes of $k \leq 1$ and the damping of $k = 2$ at $a_0 = 1$.
- The loss of amplitude of the modes next to the edge in the PIC runs is a finite-amplitude effect: it grows with ε and does not change with Δr or N . With 4×10^4 particles the mode of $k = 1.25$ persists up to $t = 4000$ at ω_d . With 10^5 particles and $\varepsilon = 0.03$ its amplitude stays within 0.90–1.14 of the linear one and its frequency is 0.15314, against 0.15326 from the loop gain.
- The reference runs depart from stationarity for two reasons: the error of order Δr^2 of the PIC field sets the initial level, and the discreteness of the particles makes it grow after $t \approx 1000$. Four times more particles remove most of the growth up to $t = 4000$, and ten times more remove it.
- η does not mark the transition in this setting.

Open.

- At small mass the discrete mode of $k \leq 1$ is bound by less than 10^{-6} band widths, so a finite simulation sees only a slowly decaying edge response. The non-damping that the theorem proves is not observable there; it becomes observable only at large mass.
- The mode of $k = 1.5$ at $a_0 = 1$, bound by 1.8×10^{-4} in frequency. Deciding whether it survives in the PIC runs needs $t \gtrsim 10^4$ and the noise of $4N$ or less.
- The dip of the amplitude of $k = 1.25$. It is deeper and earlier with larger ε : 0.32 at $t \approx 630$, about 0.6 at $t \approx 1350$ and 0.90 at $t \approx 2250$ for $\varepsilon = 0.3, 0.1$ and 0.03 . The three runs differ in the number of particles, so the law in ε is not established; runs with $\varepsilon = 0.1$ and 0.01 on the particles of series 5, with its reference, would give it.
- The noise floor at small mass, which limits the comparisons to $\sim 10^3$ time units with 10^4 particles. Lowering it needs more particles; at $a_0 = 1$, 4×10^4 particles extend the comparison to $t = 4000$.

5 Reproduction

Everything is produced by `reproducir/scripts/hadzic.py` from that directory:

step	what it does	time
lineal	the 36 equilibria, λ_{edge} (<code>exe/hadzic/lineal/lambda.txt</code>), η (<code>eta.txt</code>) and the linear solutions of the runs	40 min
preparar	initial states and <code>.par</code> files in <code>reproducir/corridas/13_hadzic/</code>	10 min
correr	the 36 PIC runs, one after another, with four threads	5.5 h
serie	h_1 and $\delta\Phi$ of the named runs, with the snapshots split among several processes; used for the two runs of series 5	1 h
analizar	h_1 and $\delta\Phi$ in the map of the equilibrium, envelopes, slopes, pole fits and the table of series 4 and 5 (<code>exe/hadzic/resumen.txt</code>)	3 h
figuras	the figures of this report	1 min
videos	one video per perturbed run, in <code>exe/hadzic/videos/</code> : the particles in (r, p_r) , the harmonic $k = 1$ of the perturbation in (Q, J) and $ h_1 $	3 h

The runs of series 1 used the Courant factor 2 of the template, so they reached $t = 8000$; their `.par` files are kept as used, and their header comment says $t_{\text{fin}} = 4000$. The runs with $\Delta r/2$ use the initial states of series 2; the dictionary CAMBIOS of the script lists the changes of each run of series 4 and 5.

References

- [HRSS] M. Hadžić, G. Rein, M. Schrecker, C. Straub, *Damping versus oscillations for a gravitational Vlasov–Poisson system*, Arch. Ration. Mech. Anal. 249:45 (2025), [arXiv:2301.07662](#).
- [HRS] M. Hadžić, G. Rein, C. Straub, *On the existence of linearly oscillating galaxies*, Arch. Ration. Mech. Anal. 243 (2022), [arXiv:2102.11672](#).
- [S] C. Straub, *Numerical experiments on stationary, oscillating, and damped spherical galaxy models*, [arXiv:2405.01235](#) (2024).

The other works of the group and the comparison of their setting with ours are in `docs/hadzic/hadzic_base.md`.